

PRANAV SATHEESH

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RESEARCH INTERESTS

Supermassive black hole dynamics and mergers, Galaxy evolution, Gravitational waves, Black hole formation, Cosmological simulations, Multi-messenger astrophysics

EDUCATION

University of Florida <i>Ph.D. in Physics</i> Advisor: Dr. Laura Blecha	2022 - Present
Indian Institute of Technology Madras <i>BS-MS in Physics</i>	2017 - 2022

APPOINTMENTS

Research Analyst Center for Computational Astrophysics Flatiron Institute, New York	2026-2027
Graduate Research Assistant University of Florida Department of Physics	2023 - Present
Graduate Teaching Assistant University of Florida Department of Physics	2022 - 2023

PUBLICATIONS

- **Satheesh, P.**, Bhowmick, A. & Blecha, L. *Galaxy merger-driven signatures in MBH pair hosts from cosmological simulations*, 2026 (in preparation)
- **Satheesh, P.**, Blecha, L., & Kelley, L. Z. (2025). *Mergers and Recoil in Triple Massive Black Hole Systems from Illustris*, *ApJ*, 993, 222 (2025), [arXiv:2506.04369](#).
- Paul, K., Maurya, A., Henry, Q., Sharma, K., **Satheesh, P.**, Divyajyoti, Kumar, P., & Mishra, C.K. (2024). *ESIGMAHM: An Eccentric, Spinning inspiral-merger-ringdown waveform model with Higher Modes for the detection and characterization of binary black holes*, *Phys. Rev. D* 111, 084074 (2025), [arXiv:2409.13866](#).

AWARDS

CCA Pre-Doctoral Fellowship by Simons Foundation	2026-2027
LSST Discovery Alliance (LSST-DA) Data Science Fellowship	2026-2028
Steigleman fellowship for graduate research in theoretical astrophysics from UF Physics	2024
Graduate fellowship for first-year graduate students from UF Physics	2022
62nd Institute day award for academic performance in physics from IIT Madras	2021
Selected for ThinkSwiss Research Scholarship	2020
INSPIRE-DST Scholarship for Higher Education	2017 - 2022

PRESENTATIONS

Invited Talks

- *What can cosmological simulations tell us about massive black hole mergers?*, **Galaxy Coffee Seminar, Max Planck Institute for Astronomy**, July 2026
- *Massive black hole mergers from cosmological simulations*, **Galaxies and Cosmology Seminar, Max Planck Institute for Astrophysics**, July 2026

Conference Talks

- *Massive black hole mergers in cosmological simulations*, **Choir conference on New Data that Challenge Underlying Assumptions in Early Galaxy Evolution, Schoodic Institute**, July 2025
- *Studying gravitational wave and slingshot recoil of massive black holes in cosmological simulations*, **33rd Midwest Relativity meeting, University of Chicago**, November 2023

Posters

- **Satheesh, P.**, Bhowmick, A., Siwek, M., Blecha, L., & Kelley, L. *Massive black hole binaries and mergers from cosmological simulations*, **16th International LISA Symposium**, June 2026
- **Satheesh, P.**, Bhowmick, A., Siwek, M., Blecha, L., & Kelley, L. *Massive black hole mergers and host galaxy properties in cosmological simulations*, **NanoGrav Fall meeting**, November 2025
- **Satheesh, P.**, Blecha, L., & Kelley, L. *Mergers, recoil and slingshot in triple massive black hole systems, The era of binary supermassive black holes*, **Aspen Center for Physics**, February 2025
- **Satheesh, P.**, Blecha, L., & Kelley, L. *Studying gravitational wave and slingshot recoil of massive black holes in cosmological simulations*, **NANOGrav Fall collaboration meeting, University of Michigan**, October 2024
- **Satheesh, P.**, Gandhi, S., & Mishra, C.K. *Fisher analysis of eccentric binaries with higher mode frequency domain inspirals*, **LIGO-Virgo-KAGRA collaboration meeting (virtual)**, March 2022
- **Satheesh, P.**, Saha, P., Schmid, H. *A spectropolarimetric method for predicting the gravitational wave polarization of LISA verification binaries*, **237th American Astronomical Society meeting**, 2021

PROFESSIONAL MEMBERSHIPS

Associate Member, NANOGrav collaboration	2024 - Present
Graduate Member, American Astronomical Society	2023 - 2025
Graduate Member, American Physical Society	2022-2023
Member, LIGO Scientific Collaboration	2021 - 2022

PROFESSIONAL SERVICE

Referee for [A&A](#) (2025)

TECHNICAL SKILLS

Programming Languages	Python, C/C++
Software & Tools	Mathematica, LaTeX, Git, GitHub, Shell scripting, hpg
High Performance Computing	Slurm, OpenMP
Python Packages	Pandas, Matplotlib, Numpy, Scipy, Astropy, Scikit-learn
Operating Systems	Linux, Windows, macOS

TEACHING EXPERIENCE

Teaching Assistant, Code-Astro 2023 <i>Taught sessions at this software engineering workshop for astronomy supported by the Heising-Simons Foundation.</i>	Summer 2023
Teaching Assistant, Physics-1 with calculus lab <i>Introductory laboratory course for undergraduates taking Physics I with calculus at the University of Florida.</i>	2022-2023
Grader, Complex Networks <i>Graduate-level course at IIT Madras.</i>	Spring 2021

SCIENCE COMMUNICATION AND OUTREACH

Service

· Organizer, Comscicon Flagship Workshop	2025-2026
· Writer and Illustrator, ComSciWri	2025 - Present
· Author, Astrobites	2023-2025
· Organizer, Physics Graduate Community	2023-Present
· Head, Horizon: The Physics and Astronomy Club of IIT Madras	2019-2020

Science-communication pieces

· Comic on massive black hole formation <i>The Origin Story of Supermassive Black Holes, ClubSciWri</i>	2024
· Science article featured in AAS Nova <i>Twinkle twinkle trail of stars how I wonder what you are</i>	2024
· Astrobites articles written by me.	2023-2025

Talks

· <i>An illustrated guide to supermassive black hole mergers</i> Graduate Student and Postdoc Seminar at UF Physics	Feb 2026
· <i>Scripting the origin story of supermassive black holes</i> Graduate Student and Postdoc Seminar at UF Physics	Sep 2024
· <i>It takes three to tango: Massive black hole triples and the final parsec problem</i> Graduate Student Postdoc Seminar at UF Physics	Sep 2023
· <i>Python for Astronomy</i> , An Youtube lecture series offered by me as part of Horizon	Jul 2020
· <i>Relativity and Gravitation</i> , Horizon-IITM Summer School	July 2021
· <i>Analysis of Globular Clusters Using Colour-Magnitude Diagrams</i> , Shaastra IITM	Jan 2020

WORKSHOPS ATTENDED

LSST-DA Data Science Fellowship Program Session 25, skAI Institute, Chicago	2026
NANOGrav Astrophysics working group hack week, UC Berkeley	2025
NANOGrav student workshop, Michigan	2024
Comscicon flagship workshop, Boston	2024
ICTS Summer School on Gravitational Wave Astronomy	2023
School on Black Holes and Gravitational Waves, Chennai, India	2022
Physics of the Early Universe, ICTS (Online)	2020